

Conductive Fabric Diaphragm for Noise-Suppressive Headset Microphone

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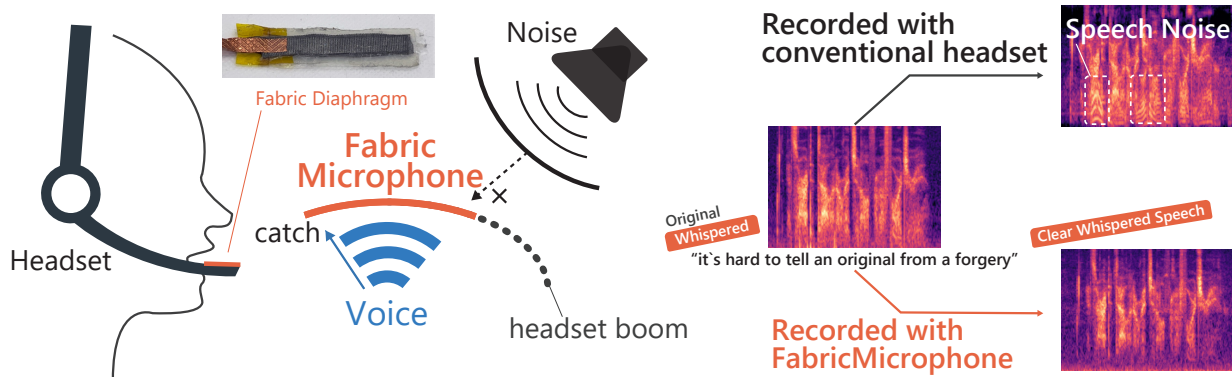


Figure 1: An overview of the headset-type fabric microphone, which can dominantly capture the user's voice.

ABSTRACT

Voice interaction systems require high-quality speech input; however, ambient noise significantly degrades the performance of conventional microphones. While multimodal sensor-based speech enhancement techniques have been proposed, they typically require GPU devices for processing. We proposed a conductive fabric diaphragm headset microphone, which integrates a planar fabric microphone into a headset configuration. The proposed microphone exhibits low sensitivity and curved directivity characteristics, enabling efficient capture of the user's voice while simultaneously suppressing background and competing speech noises.

Comparative analyses against two commercially available noise-cancelling headsets demonstrated that our approach achieved

superior Scale-Invariant Signal-to-Noise Ratio (Si-SNR) performance. Our approach constitutes a real-time, hardware-based noise-suppressive microphone solution with potential applications in speech communication, automatic speech recognition, and voice conversion technologies operating in noise-rich environments.

CCS CONCEPTS

• **Human-centered computing** → *Mobile devices*; **Sound-based input / output**.

KEYWORDS

noise-suppressive microphone, speech enhancement, fabric microphone, wearable device, voice user interface

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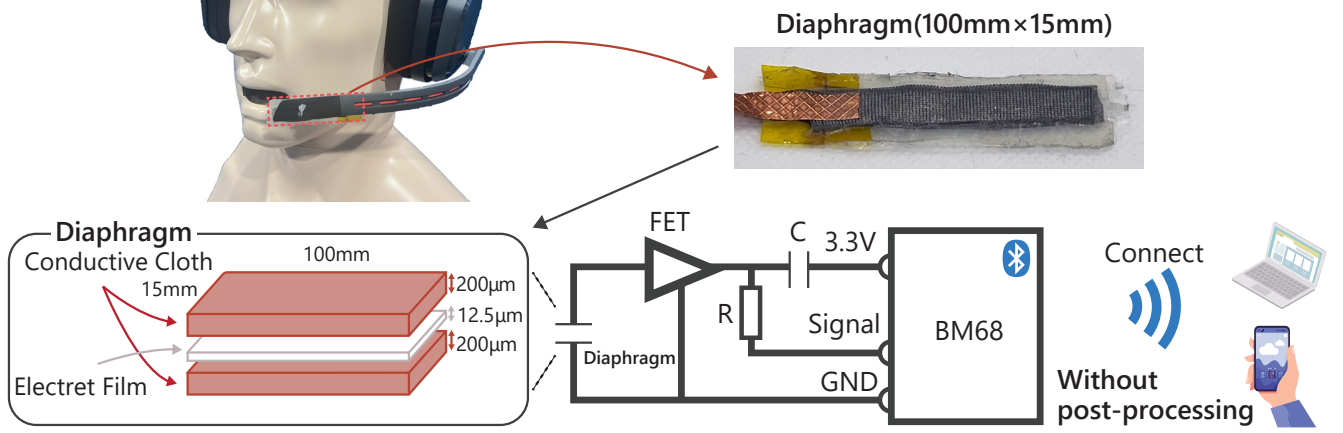


Figure 2: (a) Device: we implemented a detachable noise-suppressive microphone with a piezoelectric sensor. (b)(c) we use natural and whispered speech data from wtimit dataset [11] for simulating the user’s speech. Also, we used speech and noise clips from LibriMix [5] and Wham[18] for simulating noisy environment.

Clear speech input is crucial for various voice-based applications, such as voice communication, voice control, and voice interfaces for smart devices. However, capturing the user’s voice accurately in noisy environments with background noise or interfering speech remains a significant challenge for conventional microphones. This issue severely degrades the performance of speech recognition systems and voice-based user interfaces, leading to frustrating user experiences.

Traditional software-based methods [2–4], often rely on prior knowledge of noise characteristics, which is usually unavailable in real-world scenarios. More advanced deep learning-based approaches [13, 15, 16] have been introduced to alleviate this reliance on prior knowledge. However, these methods still suffer from the label permutation problem, where the system may mistakenly output the competing speaker’s voice as the target speech.

To address this issue, researchers have explored hardware-based solutions using multi-channel and multi-modal approaches [6, 7, 14, 17, 19]. These approaches are based on deep learning algorithms and have limited use in devices with GPUs.

Earphone-based methods, like ClearBuds [1], offer a more stable and hands-free experience by utilizing beamforming techniques to enhance the user’s speech. However, these methods struggle to distinguish between the user’s voice and competing speech coming from the front direction, as the beamforming approach is unable to differentiate between these two sources effectively.

On the other hand, wearable mask-type [8] was proposed to capture the user’s voice, especially whispered speech, in noisy environments up to 80 dB. This mask-type design may hinder natural facial expressions during conversations.

we propose a fabric-based diaphragm headset microphone, inspired by the concepts of WhisperMask[8], addressing its limitations by integrating a planar fabric microphone into a conventional headset. Fabric diaphragm has a low sensitivity and the curved diaphragm makes curved directivity, then this fabric microphone can capture only user’s voice.

1 PROTOTYPING AND EVALUATION

We developed a prototype fabric-based diaphragm headset microphone. The microphone’s diaphragm comprises a conductive fabric electrode and a dielectric plastic film (perfluoroalkoxy polymer, PFA). The voltage is amplified using FETs and connected to a BM83 audio SoC. We created two diaphragm sizes: 100cm×15mm (FabricMic 100mm) and 50mm×15mm (FabricMic 50mm).

Table 1: Mean and Standard Deviation of SiSNR for Different Microphones in Noisy Environment

Microphone	Condition	SiSNR (dB)	
		Mean	Std Dev
FabricMic 50mm	Natural	-25.23	11.98
	Whisper	-23.57	11.29
FabricMic 100mm	Natural	-24.77	12.30
	Whisper	-24.16	11.73
Jabra	Natural	-38.26	12.92
	Whisper	-50.26	12.01
Logitech	Natural	-44.51	9.84
	Whisper	-49.24	10.67

Table 2: Comparison of FabricMic with Other Microphones for SiSNR in Noisy Environment (Natural Condition)

Comparison	t-statistic	p-value	Effect Size
FabricMic 50mm vs. Jabra	7.3992	3.814e-12***	1.0464
FabricMic 50mm vs. Logitech	12.4360	1.293e-26***	1.7587
FabricMic 100mm vs. Jabra	7.5650	1.427e-12***	1.0698
FabricMic 100mm vs. Logitech	12.5298	6.689e-27***	1.7720

*** p < 0.001

To evaluate our microphone, we compared its noise-cancelling performance with two commercially available noise-cancelling headphones (Logitech[12] and Jabra[10]). For voice collection, we employed a voice simulator conforming to human mouth standards [9]. To simulate noisy environments, we generated speech noise and ambient sounds from the speech separation dataset [5, 18]. Two noise speakers were positioned 300mm from the voice simulator, while an ambient noise speaker was placed 1m away. We utilized 100 voice samples from the WTIMIT dataset [11], where the same speaker uttered identical content in both natural and whispered speech. These recordings were conducted in a quiet, non-reverberant environment with an ambient noise level of 29.8 dB.

2 RESULTS AND CONCLUSION

We conducted a comparative analysis of our two fabric microphones against two noise-suppressive headset microphones in both quiet and noisy environments. The evaluation was based on the metrics of Scale-Invariant Signal-to-Noise Ratio (Si-SNR) which calculates the similarity between clear and noisy voices. Table 1 presents the results of Si-SNR and Table 2 shows the comparison of our headsets and commercial ones.

The analysis of Si-SNR reveals significant differences in microphone performance under noisy conditions. FabricMic microphones consistently outperformed competitors, demonstrating superior noise suppression capabilities. The FabricMic 50mm and 100mm models exhibited mean Si-SNR values ranging from -25.23 dB to -23.57 dB, substantially higher than those of Jabra (-50.26 dB to -38.26 dB) and Logitech (-49.24 dB to -44.51 dB). Notably, FabricMic maintained consistent performance across both natural and whispered speech, with minimal variation between conditions. In contrast, Jabra and Logitech showed marked degradation in whisper mode, with Jabra's Si-SNR dropping by approximately 12 dB from natural to whispered speech. Statistical analysis corroborates these findings, with all comparisons between FabricMic and competitors yielding p-values < 0.001 and large effect sizes (> 1.0). The FabricMic 50mm versus Logitech comparison in natural speech conditions, for instance, showed a t-statistic of 12.4360 and an effect size of 1.7587, underscoring the magnitude of FabricMic's performance advantage.

These results suggest that our microphones are particularly effective in preserving speech signals in noisy environments, potentially offering significant benefits in applications requiring clear audio capture under challenging acoustic conditions. However, it is important to note that while Si-SNR provides valuable insights into signal preservation, a comprehensive evaluation should include additional metrics to fully assess speech quality and intelligibility.

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